



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SAKTHI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

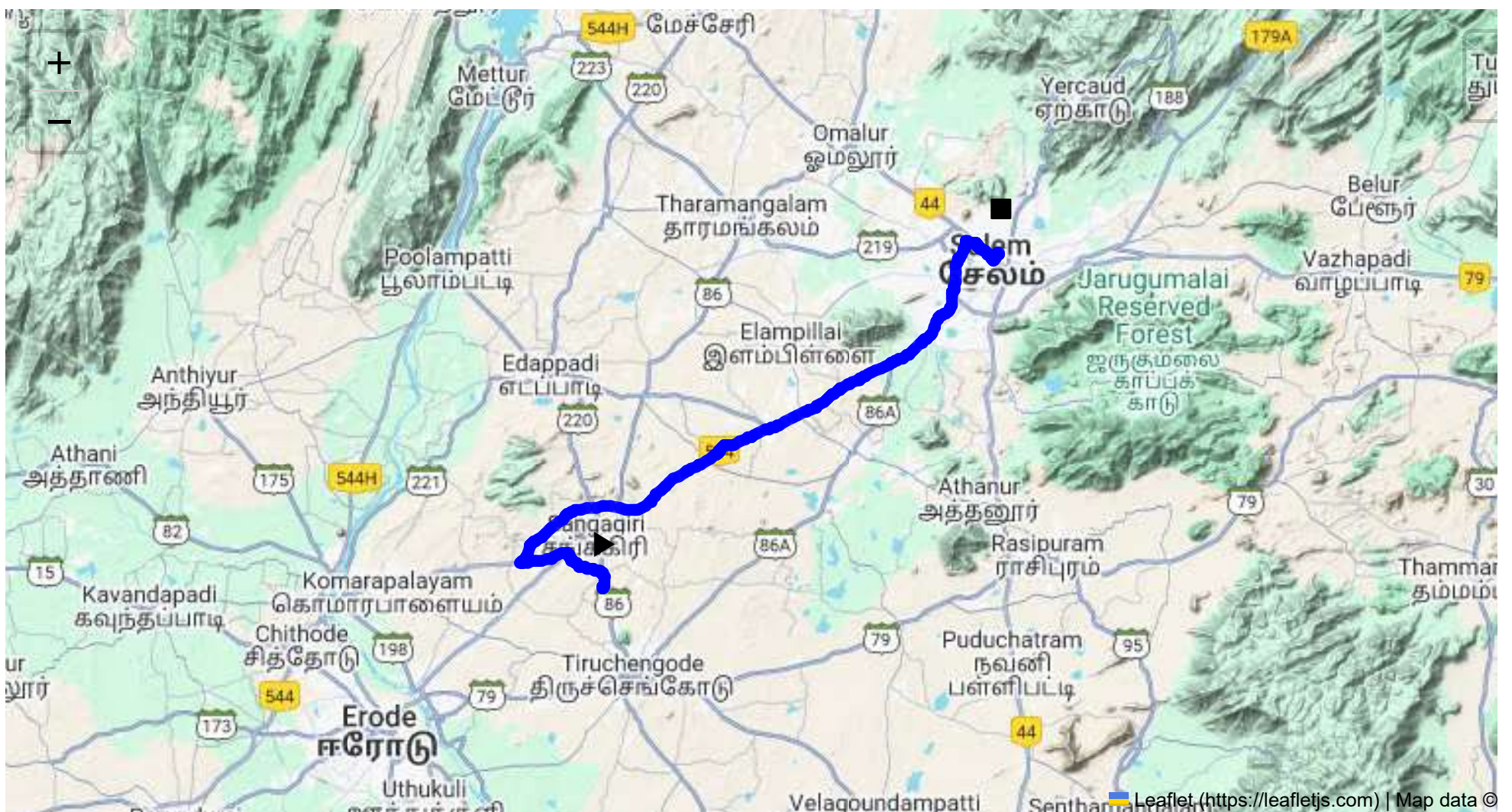
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 57.06 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.667776, 78.150671)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Salem spans approximately 57.06 kilometers, passing through Sankari and Padaiveedu along Bhavani Main Road. The journey typically takes about 1.17 hours for heavy vehicles carrying hazardous materials. The road conditions include both rural and semi-urban environments, featuring a mix of state highways and local roads.

Weather Conditions

The typical weather conditions in Tamil Nadu include hot and humid summers, a moderate monsoon season, and mildly cool winters. Potential weather-related hazards include heavy monsoon rains between June and September, which can lead to flooding and poor visibility. It's crucial to monitor weather forecasts regularly during the monsoon period to anticipate any road closures or difficult driving conditions.

Traffic Patterns

Traffic peaks during morning (7:30-9:30 AM) and evening (5:30-7:30 PM) hours when locals commute to and from work or school. The Sankari region is particularly congestion-prone during these times. Truck drivers should anticipate possible delays and plan their journey to avoid these hours when feasible.

Road Quality and Infrastructure

The road quality on this route varies. The state highways are generally well-maintained, though certain segments, particularly in more rural areas, may be narrow with unclear lane markings. Potholes and uneven surfaces might be encountered, especially after the monsoon season. It's advisable to maintain moderate speeds and exercise caution on less well-maintained roads.

Alternative Routes for Emergencies

In case of major congestion or road closure, consider detouring via NH544 at Sankari, which offers a more direct but potentially longer route to Salem. Diversions through local roads (if feasible) should be done with caution and adequate prior research due to possible inferior road conditions.

Local Regulations

Transporting hazardous materials requires compliance with Tamil Nadu's specific regulations concerning route permits, safety procedures, and vehicle marking. Ensure all documentation is up-to-date, and drivers are trained in handling hazardous materials safely.

Historical Incidents

The route has a moderate history of incidents involving heavy vehicles, primarily due to driver fatigue or mechanical failures. No significant hazardous material spills have been recorded recently, but continuous vigilance and adherence to safety guidelines are mandatory.

Environmental Considerations

The route traverses agricultural lands and a few ecological zones. Minimizing noise and emissions is crucial to prevent potential negative impacts on local flora and fauna. Be wary of livestock crossings, particularly in rural segments.

Communication Coverage

While the route is generally covered by mobile networks, certain rural areas near Sankari West might experience weak signals or dead zones. It is advisable to ensure communication devices are fully charged and emergency measures are in place.

Emergency Response Times

- Sangagiri to Sankari:** Approximately 20-30 minutes due to the proximity to local health centers.
- Sankari to Sanakar West:** May take 30-45 minutes given longer distances to the nearest facilities.
- Sankar West to Salem:** An estimated 15-25 minutes as Salem hosts numerous hospitals and emergency services.

Overall Risk Assessment

This route, while predominantly safe, poses moderate risks associated with heavy rainfall, traffic congestion, and variable road conditions. To mitigate these risks, optimal scheduling, regular vehicle maintenance, and comprehensive driver training on safety protocols should be sustained. Enhanced awareness of route-specific challenges is essential for safe transportation of hazardous materials along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	Medium	11.45352, 77.85698	30 KM/Hr
5	Turn	Medium	11.46048, 77.85036	30 KM/Hr
6	Turn	Medium	11.46303, 77.84945	30 KM/Hr
7	Turn	Medium	11.46318, 77.84913	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45634, 77.81673	15 KM/Hr
10	Turn	Medium	11.61748, 78.10604	30 KM/Hr
11	Turn	High	11.67074, 78.12668	15 KM/Hr
12	Turn	High	11.67081, 78.12661	15 KM/Hr
13	Turn	High	11.67582, 78.12786	15 KM/Hr
14	Blind Spot	Blind Spot	11.66327, 78.14760	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	Medium	11.66770, 78.14955	30 KM/Hr
16	Turn	Medium	11.66776, 78.14973	30 KM/Hr
17	Blind Spot	Blind Spot	11.66758, 78.15081	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
15	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Venkatesh Hospital and Urology Centre	11.6759517, 78.1324029	30 km/h	Medium
20	hospital	R.K. Hospital, Salem	11.6746905, 78.1339717	30 km/h	Medium
21	hospital	Kurinji Hospital, Salem	11.6771806, 78.1351224	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	hospital	Bhavani Hospital	11.6745697, 78.1362683	30 km/h	Medium
23	clinic	Ayush Care Siddha Centre	11.6731195, 78.1351967	30 km/h	Medium
24	clinic	Bismillah Piles Clinic	11.6755285, 78.1343325	30 km/h	Medium
25	clinic	Arunagiri Ent Nursing Home	11.6762823, 78.1369593	30 km/h	Medium
26	clinic	Srimaya Child Health Centre	11.6750852, 78.1370646	30 km/h	Medium
27	hospital	Sri Gokulam Hospital Pvt Ltd	11.6725694, 78.1337169	30 km/h	Medium
28	clinic	Dr. Kanaks Skin Hair & Cosmetology Clinic	11.6722777, 78.1375557	30 km/h	Medium
29	hospital	Rainbow Hospital	11.6722366, 78.13774	30 km/h	Medium
30	hospital	Gl Hospital	11.6742597, 78.1379092	30 km/h	Medium
31	hospital	Balaji Skin Hospital	11.676166, 78.1379488	30 km/h	Medium
32	hospital	Dr. Sasi Eye Care	11.6743698, 78.1379234	30 km/h	Medium
33	hospital	Dr Agarwal Vasns Eye Hospital, Salem	11.6700596, 78.1381854	30 km/h	Medium
34	police	Palapatti Police Station	11.669908, 78.1395206	30 km/h	Medium
35	police	Palapatti B 10 Police Station	11.6699556, 78.1394849	30 km/h	Medium
36	hospital	S S Medical Centre	11.6706585, 78.1398415	30 km/h	Medium
37	hospital	Saravana E.N.T Hospital	11.6720221, 78.1394629	30 km/h	Medium
38	hospital	Dr Sudhakar Ent Care Centre	11.6713519, 78.1399383	30 km/h	Medium
39	hospital	Kattabomma Hospital	11.6689589, 78.1371173	30 km/h	Medium
40	hospital	Ark Hospital	11.6716723, 78.1403218	30 km/h	Medium
41	hospital	Vasan Eye Care Hospital, Salem	11.6702294, 78.1407532	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
42	hospital	Pathanjali Ayurvedha Hospital	11.671244, 78.141525	30 km/h	Medium
43	hospital	Lotus Eye Care Hospital, Salem	11.6689406, 78.1425789	30 km/h	Medium
44	hospital	Pranav Hospitals Pvt Ltd	11.6693372, 78.142532	30 km/h	Medium
45	hospital	RSP Siddha and Varma Hospital	11.6727719, 78.1418429	30 km/h	Medium
46	hospital	Dr. Agarwal Vasans Eye Hospital	11.6696681, 78.1417073	30 km/h	Medium
47	hospital	Sudha Hospital	11.6691694, 78.142588	30 km/h	Medium
48	hospital	Dr. S. Rajendran Children Hospital	11.6707079, 78.1421878	30 km/h	Medium
49	hospital	Eka Hospital	11.6691926, 78.1425812	30 km/h	Medium
50	clinic	Salem Child Care Centre	11.6706006, 78.1421572	30 km/h	Medium
51	clinic	JH Cardiac Centre	11.6707391, 78.1421402	30 km/h	Medium
52	hospital	S K S Hospital India Pvt Ltd	11.6701839, 78.1433663	30 km/h	Medium
53	hospital	KS Siddha & Varma Hospital	11.666952, 78.141606	30 km/h	Medium
54	clinic	Homeopathy Clinic	11.6691556, 78.1434845	30 km/h	Medium
55	clinic	K.S. Siddha & Varma Clinic	11.6674481, 78.1392422	30 km/h	Medium
56	hospital	S.K.S. Hospital	11.670548, 78.143109	30 km/h	Medium
57	hospital	Sri Vaari Raja Sidha Hospital	11.6668535, 78.1422245	30 km/h	Medium
58	clinic	Salem Poly Clinic	11.6663605, 78.144658	30 km/h	Medium
59	hospital	S.P.S. Hospital	11.6669469, 78.1445952	30 km/h	Medium
60	clinic	Abhi Eye Hospital	11.6641165, 78.1457788	30 km/h	Medium
61	hospital	Krishna Fertility and Laparoscopy Hospital	11.6643332, 78.1459989	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
62	hospital	Konlampatty Enocall Hospital	11.6642816, 78.1459647	30 km/h	Medium
63	hospital	St.Marys Hospital, Salem	11.6634501, 78.1444402	30 km/h	Medium
64	hospital	Nithish Ge And Multi Speciality Hospital	11.6636554, 78.1457305	30 km/h	Medium
65	hospital	Satya Sai Ent Head and Neck Hospital	11.664806, 78.1466813	30 km/h	Medium
66	clinic	Dr. S. S. Meera Vasudevan Clinic	11.6634071, 78.1457976	30 km/h	Medium
67	clinic	Sri Dhanvanthari Siddha Clinic	11.6634617, 78.1477869	30 km/h	Medium
69	clinic	Dr. Rajenderan Child Clinic	11.6643448, 78.1483264	30 km/h	Medium
70	clinic	Kathirveel Orthopeadic Hospital	11.6621451, 78.1473041	30 km/h	Medium
71	hospital	Prn Critical Care Centre And Hospitals	11.666386, 78.1493931	30 km/h	Medium
73	clinic	Omega Clinic	11.660922, 78.150292	30 km/h	Medium
74	hospital	Vijaya hospital	11.6674308, 78.1505122	30 km/h	Medium
75	hospital	Vijaya Hospital, Salem	11.6678019, 78.1506725	30 km/h	Medium
76	clinic	Sugam Homoeo Clinic	11.6655186, 78.1518049	30 km/h	Medium
77	hospital	Andandhammal Hospital	11.669873, 78.14765	30 km/h	Medium
78	clinic	Aravind Vision Centre	11.670354, 78.151069	30 km/h	Medium
79	hospital	Revathi Hospital	11.6698571, 78.1526859	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
16	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
68	school	St. Mary's Hr. Sec. School	11.662587, 78.1449627	30 km/h	Medium
72	school	Little Flower School	11.662629, 78.1495633	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61748, 78.10604



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67074, 78.12668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67081, 78.12661



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67582, 78.12786



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.66327, 78.14760



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66770, 78.14955



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66776, 78.14973



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.66758, 78.15081

